SENTIMENT BASED STOCK MARKET PREDICTION

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Abstract — **The Stock Market prophets focus on developing a efficient approach to predict the future stock prices. Apart from achieving best results in the successful stock market prediction it is also necessary to minimize the inaccurate forecast of stock prices. The project aims to design and implement a predictive system for guiding stock market investiture. As stock market is a very volatile environment, it can be influenced by positive or negative stock price releases. The research shows that there is an effect of prices on stock market and that the stocks can be predicted with the help of those prices. Automated data in general and company's distinct data of Apple, Microsoft, Google and Amazon are used to test the effect of the efficient prices on the stock market. A different procedure is experimented with sentiment analysis and the results display that it is able to predict the stock market with the help of stock prices for different companies, where the linear regression model is possible to predict more accurately than the sentiment analysis method.**

Keywords — Prophets, predictive, investiture, sentiment analysis, linear regression.

I. INTRODUCTION

Stock market is an important division of the economy of a country. Complex behavior and unstable nature are difficult and essential tasks for the stock market decision. As tremendous amount of the data is produced by stocks each day, it is difficult task for an individual to analyze every past and present stock price information for predicting the future trend of stocks. Stock trends are affected by many factors, and one of which is daily stock price articles. Daily stock news articles can play an vital role for prophets while judging the stock prices. Hence, it is become necessary to efficiently analyze the information to hold up and assist the investors to make smart trading resolution before making the real investments.

A. Stock Market Analysis

Stock market is an essential part of a country because to increase the funds for future expansion or set up a new business, company has to either take a loan from a monetary organization or it has to distribute shares through the stock market. In order to raise the funds for business expansion, stock market is the primary source for any company. It plays an important character on the common platform for buyers and sellers of the stocks at the stock market. It is also pretentious by investor sentiment or conviction, which can cause the market to go high or low, which can cause stock prices to rise or fall in the market.

The stock trend uses sentiment analysis for the dataset of four companies (Apple, Google, Microsoft and Amazon) along with the price of stock to predict the future behavior of the stock market. Here the technique used is that, initially the stock news articles of company are fetched from business website. The stock articles are classified into positive, negative and neutral based on the sentiment of the stock price using Machine learning algorithm like support vector classifier and the technique used is Linear Regression model. This sentiment analysis on classification is used to determine the future trend of a stock market. If the positive sentiment is found, there is more probability that price of the stock will increase further and if negative sentiment is found, then price of the stock may decrease.

II. RELATED WORK

Stock market prediction has been an vital area of research for a long time. Stock prices are stated by Efficient Market Hypothesis (EMH) that are largely driven by new information and is widely accepted by the research community. In general, stock markets are governed by central paradigm and several people have attempted to extract efficient patterns that stock markets behave and respond to external stimuli.

The correlation between "Public sentiment and market sentiment" [6] analyzes on public available Twitter data v/s stock data to find public mood into 3 classes like Positive, Negative, Neutral. Research has proved that these three measure classes are highly correlated and give similar results in identifying most important stock prices in the market. Hence it results in better correlation between calm and happy mood dimensions from dataset values.

The designed feature matrix for stock market prediction [10] which describes the correlation between "Twitter sentiment and stock prices". It classifies whether the business and financial news is positive, negative or neutral and hence represent

the stock trend of various companies. The implemented model works better on sentimental analysis with 75% accuracy.

The correlation between "Stock market values and sentiment in Rss news feeds" [9] which the classification is done in word/phrase level, sentence level in 3 classes. The Research stock market values combines the sentiments of people through the news feeds and senses data to predict the behaviour of stock.

Research evaluates the correlation between "Stock market values and sentiment in tweets"[11]. The collective mood of public derived from twitter feeds describes the strong correlation exits between rise/fall in stock prices to the public opinions expressed on tweets. Hence it evaluates the past data pertaining to the stock market and its prices of the corresponding time period, in order to make predictions in stock trends.

III. PROPOSED WORK

Before implementation of any system, it requires to be examined for its flaws. A blueprint diagram of the complete system comes into mind for examine the flaws. The proposed blueprint must be easy enough for a amateur to perceive, understand and detailed enough to describe all the major working modules of the system.

The proposed system shown in Figure 1 comprises of loading the dataset of different companies (Apple, Google, Microsoft and Amazon) which helps the user to load into the system. After the successful extraction of the dataset, user can select a company from the list of companies like Apple, Amazon, Microsoft and Google. Once the company is selected, stock news related to that company is fetched from the business website. The fetched stock news is stored in a file which is then pre-processed using the model called Linear Regression model. As the data pre-processed is unsupervised, in order to train the data of the respective company it is converted to the supervised data by assigning the labels.

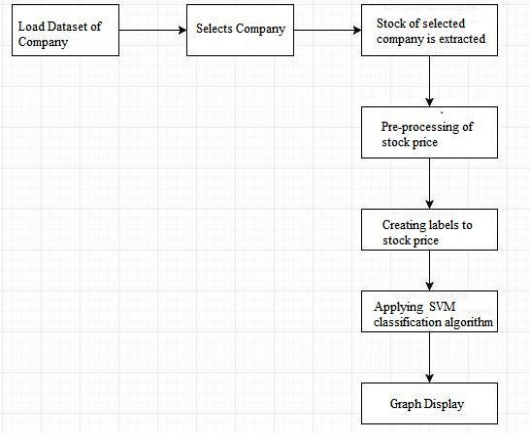


Figure 1: Description of Proposed System

Then stock news are trained to SVM classification algorithm by using these labels. The SVM classifier classifies the news as positive, negative or neutral. Lastly graph of the changing stock trend is displayed along with the sentiment obtained.

IV. METHODOLOGY

To achieve the goal of proposed work, system defines the architecture, modules, interfaces and data for a system to satisfy the stated requirements. System design will be seen as the applications of system theory to product development. System analysis, system architecture and system engineering are the disciplines introduced by the system design. It describes how the system works and overall architecture.

Architecture model in Figure 2 represents the main components of the system. Each components constitutes the working of system and together gives functionality of the system.

A. Physical view of system

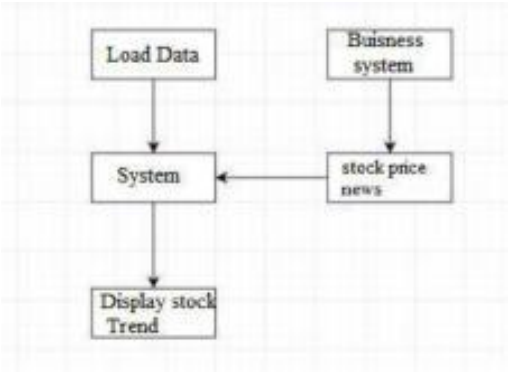


Figure 2: Physical view of system

1) Load Company Data: The Dataset is taken from web source for different companies like Apple, Google, Microsoft, and Amazon and is loaded to jupyter notebook for pre-processing.

2) System: Stock trend system which process the stock price content and exhibits the changing stock trend along with sentiment of that price news.

3) Business system: Another system from which news related to stock are fetched for each company.

B. Process view of system

Architecture model represents the main process of the system as shown in Figure3.

The system starts with the extraction of dataset for corresponding company (Apple, Microsoft, Google and Amazon) and simultaneously the dataset checks for Null values that should not contain any empty values in any of the attributes in dataset.

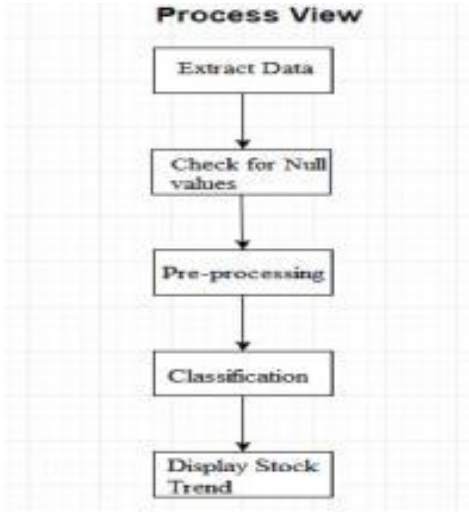


Figure 3: Process view of system

The third step involves extraction of data where extracted data are pre-processed to get important tokens from stock news. Then each sentence is classified as positive or negative or neutral using SVM classification algorithm. Lastly displaying the graph for varying stock trend of each company is plotted with stock volume along y-axis and Date along x-axis.

V. DEVELOPMENT

The subchapter gives brief explanation on the description of software system and detailed workflow. The Methodology gives the complete analysis of each process involved in Figure4.

A. Development Process

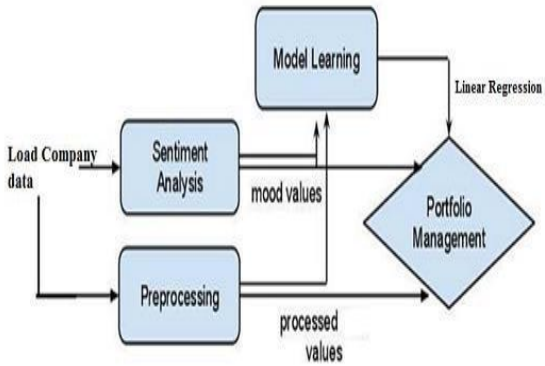


Figure 4: Block Diagram of System[6]

1) Data Pre-processing

The Data pre-processing involves four stages:

- Data Discreatization: Part of data reduction but with particular importance, especially for numerical data.
- Data Transformation: Normalization of data.
- Data Cleaning: Fill in missing values.
- Data Integration: Data files are integrated into one complete system.

After the dataset is transformed into clean dataset, it is split up into training and testing sets for further evaluation. The training values are taken as the more recent values. Testing values are kept as 10-20% (i.e. 0.2% test set and 0.8% train set) of the total dataset.

B. Feature Selection and Generation

The new features are developed from the base features which gives better insights of data like "Changing during day" and "Changing previous day" of stock price. Selection of features are made on closest scores, with the help of an linear regression for each company. Further linear regression sentiment score is added as an feature for each company depending on stock price about that specific company and also the score on that company's page. Finally graphical representation of stock trend for four companies are displayed.

C. Classification

The classification method used for our work is Support Vector Machine (SVM) classification. In machine learning, SVM's are supervised learning models which associates the learning algorithms that is used to analyze data for classification and regression analysis. Given a set of training data values, each marked as pertaining to one or the other of two categories, an SVM training algorithm builds a model that allocates new data values to one category or the other, making it a non-probabilistic binary classifier.

Support Vector Machine classifier is a supervised classifier, where input of the algorithm is stock price and labels that are created. This information is trained to SVM learning algorithm to classify the stock price as positive, negative or neutral.

Algorithm: Support Vector Machine (SVM)
Input: Training set S; Number of Base classifier C; Test instance x'.
Training Process:
For i=1:C
Use Bootstrap technology on S to generate training subset S_i;
Train the ith SVM classifier SVM_i;
label SVM-R algorithm to adjust the classification hyperplane of SVM_i to form SVM-R_i;
end

Testing Process:

For i=N: C

Put x' into classifier SVM-R_i to predict its class

label y_i' ;

end

Use majority values to get the terminal class label y'; **Output:** y' which is predicted class label for test instance x' .

The algorithm for the work flow of the system using Support Vector Machine (SVM) shows the performance of model for stock price is evaluated into positive, negative or neutral.

D. Model Learning

Simple Linear Regression: Simple Linear Regression is one of the Machine learning technique which is useful for finding relationship between two continuous variables. One is Independent variable (predictor) and other is dependent variable (response). As linear regression is a statistical relationship, two variables is said to be deterministic if one variable can be accurately expressed by the other. The major idea is to get a line that best fits the data. The best fit line is the one that calculates total prediction error (all data value points) which is as small as possible.

Company dataset which contains stock data and can be took any of the attribute in order to predict future trend of stock price. Now relationship between 'Volume' and 'Close' are taken up for linear model as our training data. Goal is to design a model that can predict the price if given close prices are studied. A regression line is obtained by using the training data which provides minimum error. Hence the input value (Close) should predict their price with least error.

$$Y(pred) = b0 + b1 * x$$

Equation [1]

The values b0 and b1 should be chosen in order to minimize the error. To evaluate the model, sum of squared error is taken as metric and achieves to obtain a line that best reduces the error. If the error is not squared, then positive and negative values will cancel out each other.

E. Analysis

For analyzing the performance and efficiency of the stock price, we used the Root Mean Square Error (RMSE) and R^2 score value.

1) Root Mean Square Error (RMSE):

RMSE measures the square root of mean/average of square of all of the error. It is very commonly used and makes an standard general purpose error metric for numerical predictions. By comparing the similar Mean Absolute error, RMSE magnifies and rectifies large errors.

$$\text{RMS ERROR} = \sqrt{\frac{\sum_{i=1}^{n} (\hat{y_i} - y_i)^2}{n}}$$

Equation [2]

The RMSE error can be plotted graphically as shown in Figure 5, by considering Independent variable at x-axis and dependent variable at y-axis.

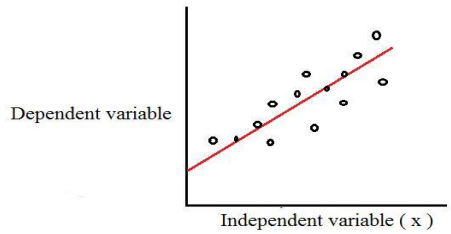


Figure 5: RMSE Error

2) R-Squared value (R^2 value)

The R^2 value can range between 0 and 1, and its higher value is more accurate for the regression model as the more variability is described by linear regression model. R^2 value refers to the proportionate amount of variation in dependent variable explained by the independent variables. It is a statistical measure that shows how close the data are to be fitted in regression line.

The Efficiency of the stock market price is analyzed using Root Mean Square Error (RMSE) and R^2 score value for each of the company's Apple, Google, Microsoft and Amazon, using linear regression model which results in 0.08278743 (8.2%), 0.31317175 (31%), 0.3618005 (36%) and 0.0593490 (5.93%) respectively. The Figure 6 shows the graphical representation of R-square error for four company's respectively, which is plotted for company along x-axis and R-square value (%) along y-axis.

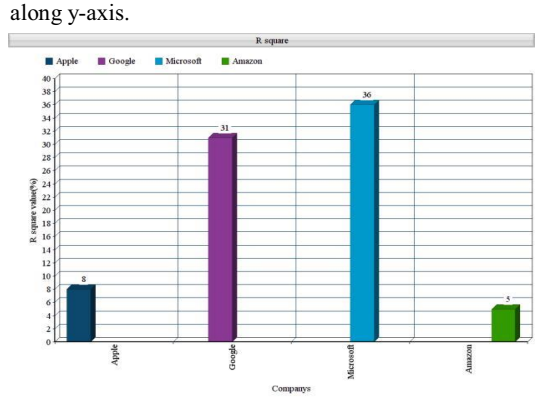


Figure 6: R-square value for each company

F. Modules:

1) Creating Labels:

After Pre-processed price is obtained, we are left with assigning labels to the stock prices. This can be achieved using one attribute as forecast column (ex: Close) and this provides labels to each stock attributes as positive or negative or neutral, then each attribute in dataset is associated with labels and the label having highest frequency or specific price is returned.

2) Classification:

After labels are created to each company, we get supervised data which is sent to the classification algorithm for learning.It provides high accuracy when compared to other algorithms. Once learning is done, the SVM algorithm classifies the stock prices as positive or negative or neutral.

VI. RESULT ANALYSIS

Based on the result obtained for variations, prediction and accuracy for the corresponding companies like Apple, Google, Microsoft and Amazon are shown below:

A. Variations of each company using"Close" independent attribute:

Variations in stock trend graph presents the clear picture of stock market price fluctuations. Closed stock price variations of each company are plotted for dates using Exponential moving averages (EMA) indicator and it considers most recent stock prices like 12 days (some day) stock price. The graph above 0 value indicates increase in stock price which tells investor to buy stocks and below 0 value indicates decrease in price which tells investors to sell the stock.

Figure 7, Figure 8, Figure 9 and Figure 10 represents the variations of stock price for Apple, Google, Microsoft and Amazon company respectively, which is plotted for dates along x-axis and close price along y-axis.

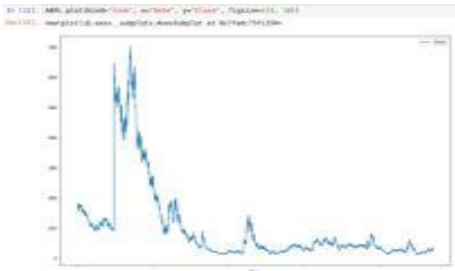
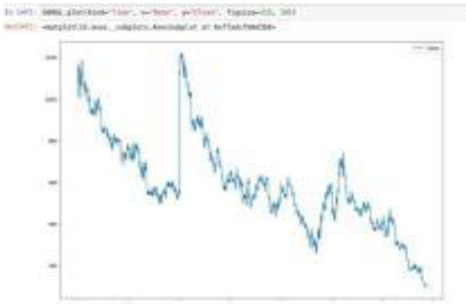


Figure 7: Apple variation

Figure 8: Google variation



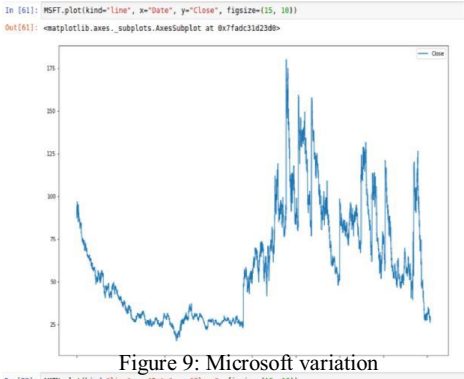


Figure 9: Microsoft variation

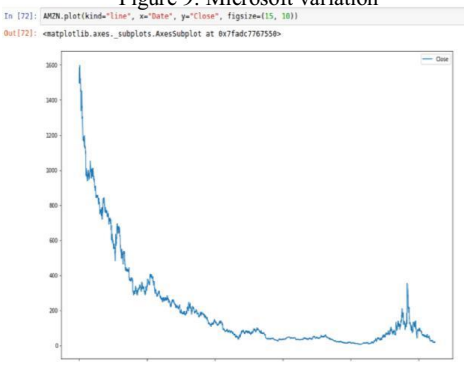


Figure 10: Amazon variation

B. Prediction of each company stock with its Accuracy:

```
Day 1 prediction: 26.5027818347
Day 2 prediction: 26.9311550506
Day 3 prediction: 27.4648570589
Day 4 prediction: 27.1063750684
Day 5 prediction: 26.3136128419
Day 6 prediction: 25.4699665616
Day 7 prediction: 23.9591609945
Day 8 prediction: 23.6417667103
Day 9 prediction: 24.998726905
Day 10 prediction: 25.7272698379
Day 11 prediction: 25.7598801559
Day 12 prediction: 25.8202092443
Day 13 prediction: 25.4143737338
Day 14 prediction: 24.9843720873
Day 15 prediction: 24.2426855221
Day 16 prediction: 24.5287296652
Day 17 prediction: 24.5495021943
Day 18 prediction: 24.4872172465
Day 19 prediction: 23.0614590811
Day 20 prediction: 23.5726262398
Day 21 prediction: 22.9674559642
Day 22 prediction: 23.4158314277
Day 23 prediction: 24.1446937309
Day 24 prediction: 24.8950687638
Day 25 prediction: 25.0627214551
Day 26 prediction: 24.765180004
Day 27 prediction: 23.8616948127
Day 28 prediction: 24.1500019447
Day 29 prediction: 23.9391054962
Day 30 prediction: 22.6457471778
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Figure 11: Prediction of stock company

The prediction of each stock company is evaluated for corresponding company and also its accuracy is measured as shown in both figure 11 and figure 12.

COMPANY	ACCURACY
APPLE	81.3488%
GOOGLE	98.2878%
MICROSOFT	98.2186%
AMAZON	98.2186%

Figure 12: Accuracy of stock company

C. Comparison of four companies on stock

Stock market trend graph in Figure 13 represents the clear picture of stock market price fluctuations for each company. The "Close" target attribute is plotted for a closed stock price variations. The results that Amazon company starts with increase in price and then decreases rapidly. Google company stock price is not decreased below 200 and no rapid changed in price of stock. Microsoft and Apple do not have much increase in their stock price and maintained constant stock after 2000.Hence Figure 20 indicates increase in price, tells prophets to buy stock and decrease in price, tells prophets to sell the stock.

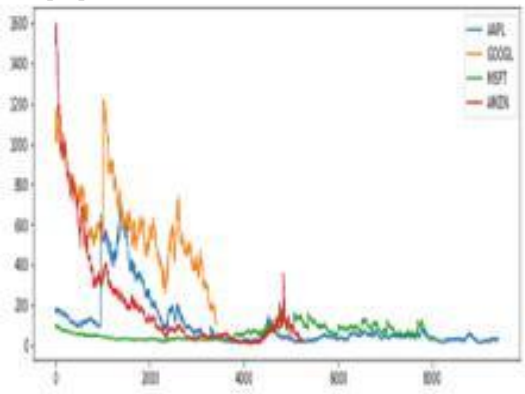

Figure 13: Comparison of four companies

VII. CONCLUSION AND FUTURE SCOPE

A. CONCLUSION

The Stock market determines the future trend of a stock by analyzing daily stock prices of four companies and classified the prices as positive or negative. After selecting a company the system fetches the dataset. Then data is processed within the system and result of the selected company is displayed. The results of proposed model are achieved higher accuracy for sentiment analysis in determining the price polarities by using SVM algorithm. Prediction and accuracy for four companies is calculated by adding more company's data. The accuracy of the Apple Company is 81.3488 %, Google Company is 98.2878 %, Microsoft Company is 98.2186 % and Amazon Company is 98.2186 %.

From the Result, it can be inferred that higher the accuracy more is the precision of stock price. As, Google's company has more accuracy value compared to the Apple, Microsoft and Amazon Companies, investor can buy the Google's stock Dataset and it is efficient for the prediction accuracy.

B. FUTURE SCOPE:

In future work, a much more scope for research exits as this is a huge and trending topic and a little improvement is also more valuable. The project can be extended by adding more company's data and check the prediction accuracy. Some technical analysis indicators can also be updated and also the recognition of emotional sentences in determining news polarities, as well as the influence of news that appears in social media can be considered.

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